

Linear Algebra – MAT 2610

Section 6.1 (Inner Product, Length and Orthogonality)

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Definition

If u and v are vectors in R^n , the number $\underline{u^T v}$ is the inner product of u and v , written $u \circ v$.

It is also often called the **dot product**.

$$\begin{bmatrix} \end{bmatrix} \begin{bmatrix} \end{bmatrix} = u \circ v$$

Example

$$\text{If } u = \begin{bmatrix} 2 \\ 1 \\ -3 \\ 2 \end{bmatrix} \text{ and } v = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 4 \end{bmatrix}, \text{ then}$$

$$u \circ v = \underbrace{[2 \ 1 \ -3 \ 2]} \begin{bmatrix} 1 \\ -1 \\ 0 \\ 4 \end{bmatrix} = \underbrace{(2)(1)} + \underbrace{(1)(-1)} + \underbrace{(-3)(0)} + \underbrace{(2)(4)} = 9$$



Theorem 1

If u, v and w are vectors in R^n , c a scalar, then

1. $u \circ v = v \circ u$ 
2. $(u + v) \circ w = u \circ w + v \circ w$ 
3. $(cu) \circ v = c(u \circ v) = u \circ (cv)$ 
4. $u \circ u \geq 0$ with $u \circ u = 0$ if and only if $u = 0$ 



Definition

If v is a vector, the length (or norm) of v is denoted as $\|v\|$ and is equal to

$$\underline{\|v\| = \sqrt{v \circ v}}$$

Or

$$\underline{\|v\|^2 = v \circ v}$$



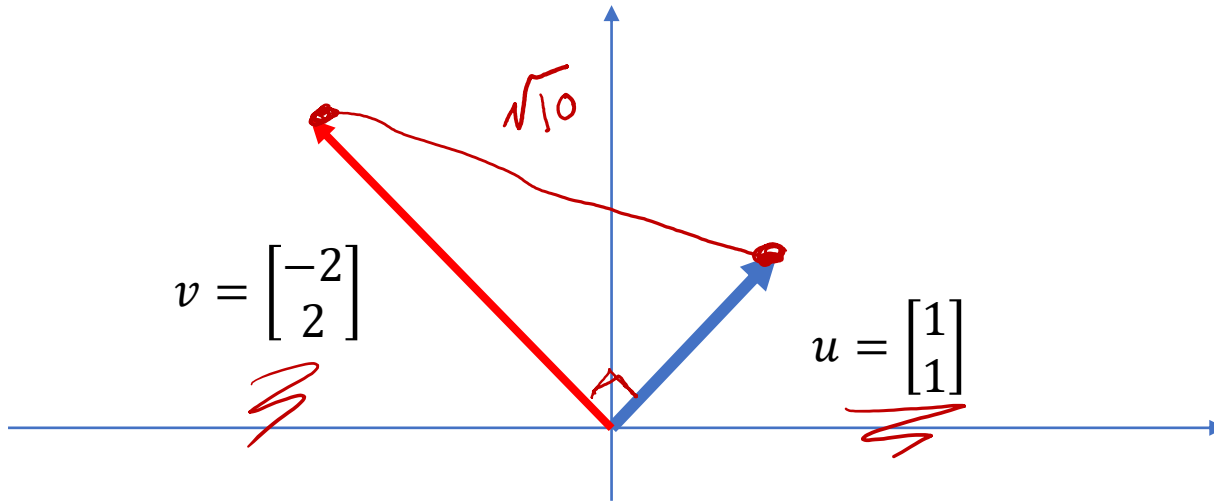
Definition

If u, v are vectors in R^n , the **distance between u and v** , denoted as $dist(u, v)$ is equal to

$$dist(u, v) = \| \underbrace{u - v} \|$$



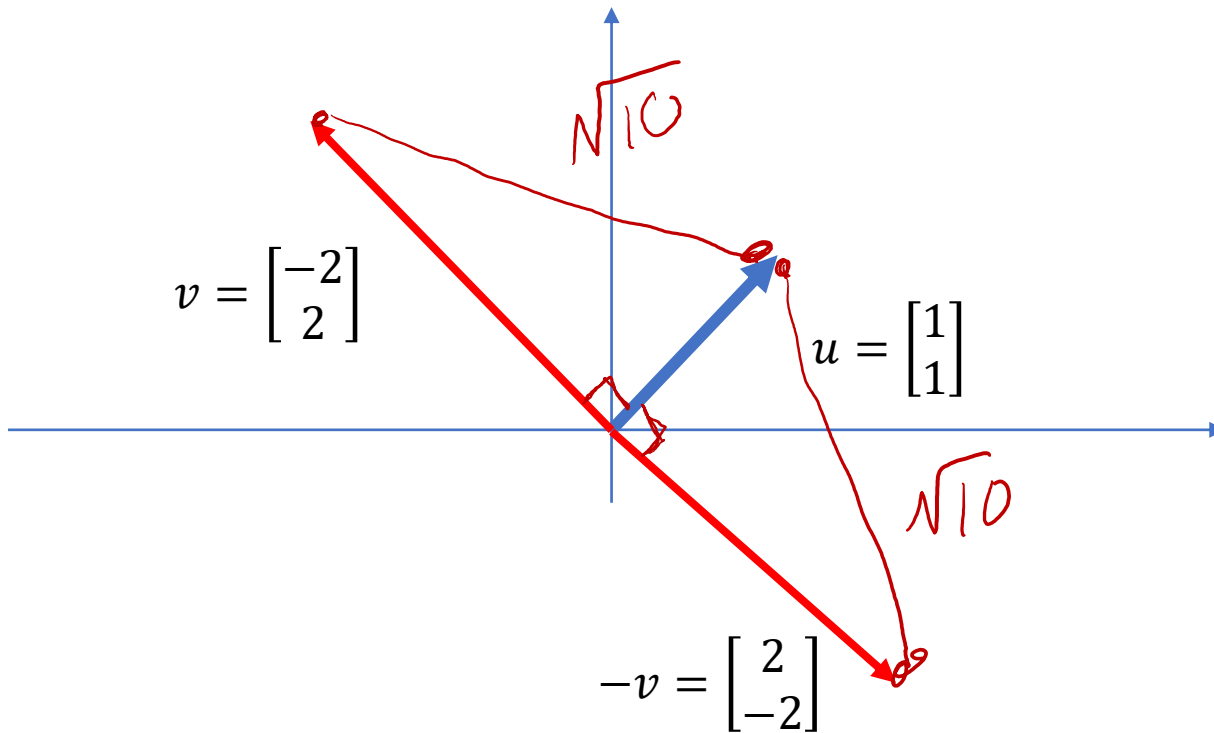
Example



$$\begin{aligned} \text{dist}(u, v) &= \|u - v\| \\ &= \left\| \begin{bmatrix} 1 - (-2) \\ 1 - 2 \end{bmatrix} \right\| \\ &= \left\| \begin{bmatrix} 3 \\ -1 \end{bmatrix} \right\| \quad \leftarrow \\ &= \sqrt{\begin{bmatrix} 3 \\ -1 \end{bmatrix} \circ \begin{bmatrix} 3 \\ -1 \end{bmatrix}} \quad \leftarrow \\ &= \sqrt{3^2 + (-1)^2} = \sqrt{10} \quad \leftarrow \end{aligned}$$



Example



Note that
 $\text{dist}(u, v) = \text{dist}(u, -v)$
because the lines are
perpendicular



$$\begin{aligned}
 \underline{[dist(u, -v)]^2} &= \|u - (-v)\|^2 = \underline{\|u + v\|^2} \\
 &= \underline{(u + v) \circ (u + v)} \\
 &= u \circ (u + v) + v \circ (u + v) \quad \leftarrow \\
 &= \underbrace{u \circ u + u \circ v + v \circ u + v \circ v}_{\leftarrow} \\
 &= \|u\|^2 + \|v\|^2 + 2(u \circ v)
 \end{aligned}$$

Similarly

$$\underline{[dist(u, v)]^2} = \|u\|^2 + \|v\|^2 - 2(u \circ v)$$

If perpendicular $(dist(u, v) = dist(u, -v))$
 $\Rightarrow 2(u \circ v) = 0$



Definition

If u, v are vectors in R^n , they are **orthogonal** if

$$\underline{u \circ v = 0}$$

$$u = (-2, 2) \quad v = (1, 1)$$

$$u \circ v = \begin{bmatrix} -2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = -2 + 2 = 0$$



Theorem 2

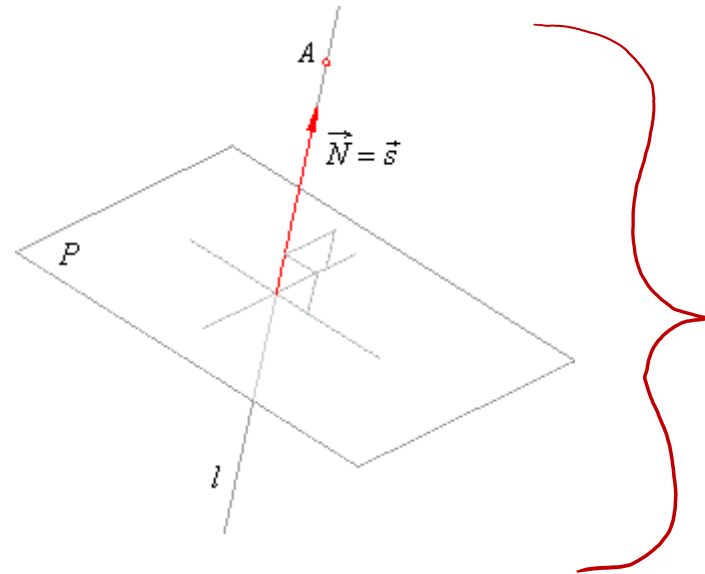
Two vectors u, v are orthogonal if and only if

$$\underbrace{\|u + v\|^2}_{\text{left side}} = \underbrace{\|u\|^2 + \|v\|^2}_{\text{right side}}$$



Definition

If W is a subspace of R^n , then the set of all vectors z that are orthogonal to every vector in W is called the **orthogonal complement** of W and is denoted $W^\perp$



Theorem 3

Let A be an $n \times n$ matrix. The orthogonal complement of the row space of A is the null space of A , and the orthogonal complement of the column space of A is the null space of A^T

$$[\text{col}(A)]^\perp = \text{null}(A)$$

$$[\text{row}(A)]^\perp = \text{null}(A^T)$$



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